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A Preliminary Study on Mitigating Upcoding in Indonesia's National Health Insurance using Multi-Agent Multi-Armed Bandits

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Abstract

Developing countries are often faced by corruption. Corruption threatens public sector organizations' viability. For instance, recent corruption at BPJS Kesehatan (Indonesia's universal health care provider) has recently led to enormous financial losses. Adversaries causing losses intelligently game the system. Existing static solutions cannot easily keep pace with the adversary. Limited audit resources further compound the problem. With hard budget and human-resource constraints, how can auditing efforts be directed to wisely choose claims that are suspicious for human review? We introduce a decentralized auditing system built with Multi-Agent Multi-Armed Bandit (MAMAB) framework. Contextual Bandits models allow us to reason about individual claims using attributes of a patient's claim. Bandits with Knapsacks (BwK) allow us to reason about the daily limit of human reviewers. Our goal is to leverage MAMAB to efficiently utilize limited resources and choose claims with the highest likelihood of containing fraud for review by a human. Moreover, we can theoretically allow regional BPJS branch offices to act as individual agents that share their knowledge of adversary behaviors (instead of sharing patients' information) to quickly learn adversarial patterns and minimize overall regret. In this paper, we present a high-level problem formulation and system design to implement MAMAB in BPJS Kesehatan's constrained environment.

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1. Introduction

Developing countries face numerous socio-economic challenges that hinder national growth, with systemic corruption remaining one of the most difficult obstacles [1]. In Indonesia, corruption can be found across various critical sectors, including banking, insurance, and healthcare [2]. Unfortunately, a major for target for this type of fraud is the Badan Penyelenggara Jaminan Sosial Kesehatan (BPJS Kesehatan) which is the state agency established to provide universal health coverage to the Indonesian population [3].

The Indonesian Ministry of Health has categorized eight main types of healthcare fraud. These include phantom billing, phantom/manipulation diagnosis, self-referrals, repeat billing, fragmentation, bribery, illegal fees, and upcoding [4]. In this study, we will focus on upcoding because it remains quite difficult to handle. This is primarily because the practice involves altering a patient's diagnosis or procedure codes to bill at a higher, medically unjustified rate. Additionally, malicious hospitals are constantly changing their tactics to avoid the detection systems. This makes it hard to detect upcoding with traditional rule-based algorithms.

For example, in July 2024, the Corruption Eradication Commission (KPK) discovered that the state had lost Rp 35 billion from upcoding at BPJS Kesehatan. This case is an obvious example of how the ongoing fraudulent claims are threatening the financial stability of BPJS Kesehatan [5]. The KPK also stated that of the 4,341 BPJS Health claims they investigated, they found 30,000 entries that were inconsistent or might have been fraudulent, and 40 percent had markups as high as 200% above the normal price. This ongoing case ultimately cost BPJS Rp 20 trillion, thus severely threatening its financial stability [6].

To overcome this problem, we adopt Multi-Armed Bandits (MAB), which is an artificial intelligence approach capable of optimally making a series of decisions under uncertain circumstances [7]. However, a standard MAB approach is not enough since the probability of fraud is deeply tied to the specific attributes of a patient and their visit. Instead, we rely on Contextual Bandits, which is an extension of the framework where the system evaluates the unique context of every claim to accurately adapt and identify fraudulent behavior over time [7].

Nevertheless, BPJS branch offices cannot audit all questionable claims since human auditors are highly limited. For this reason, we add the Bandits with Knapsacks (BwK) framework to our system. Rather than simply copying the idea of the classical knapsack problem, BwK turns the whole system against itself and forces it to be very thoughtful. The system must foresee changes and select a certain group of claims that will maximize detecting fraud. More importantly, it does all of this while making sure the required effort never exceeds what a human auditor can handle in a single day [7].

Since BPJS operates on a big scale and spread out throughout Indonesia, we need a solution that fits this reality. Therefore, we use a multi-agent system (MAS) which is defined as a system where several autonomous agents perceive their environment, behave accordingly, and interact with one another [8]. Together with these concepts, we propose a "Multi-Agent Multi-Armed Bandit" (MAMAB) architecture.

The MAMAB model proposed here involves multiple decentralized agents (the regional BPJS agencies) working in parallel and solving the same cBwK problems. This collaborative learning system allows agents to help each other by sharing fraud patterns (mathematical weights). For example, an over-coding method which an agent in Jakarta found can also be recognized by an agent in Surabaya. Hence, this will speed up detecting best audit targets and reduce collective regret very significantly.

1.1. Related Work

Traditionally, the approach to fraud prevention in the healthcare and financial sectors was predominantly focused on changing the rules of the organizations, strengthening the control system inside the organization, and training the staff extensively [9]. On the other hand, as fraud techniques evolve in complexity, fraud detection strategies are increasingly relying on supervised Machine Learning (ML) algorithms. Recently, studies comparing different ML models on imbalanced transaction data have illustrated the compromises involved with the use of standard ML models [10]. For example, Logistic Regression is a very interpretable and easily implementable model, but it is not very flexible in terms of feature interactions and nonlinear decision boundaries. Decision Trees, on the other hand, can capture nonlinear behavior; however, single trees tend to overfit the training data when there is a big imbalance in the class distribution. Ensemble methods such as Random Forest and Light Gradient Boosting Machine (LightGBM) not

only control overfitting better but also provide more stable models and less memory usage. Deep learning models such as Gated Recurrent Units (GRU) have the capability to derive sequential or contextual representations from categorical fields but are also likely to have a relatively high false-positive rate [10]. To combine the advantages of these different methods, cost-sensitive ML frameworks that emphasize high recall by adjusting decision thresholds considering the cost factor have been successfully implemented making it possible to catch 98% of the actual fraud by abandoning static, unchanging rules [11].

Traditional supervised learning models perform well when trained on historical data. Despite that, their performance often declines in real-world environments because malicious behavior changes over time. A problem known as *concept drift* [12]. As a result, researchers have recently explored the use of Multi-Armed Bandit (MAB) algorithms for adaptive fraud detection. Earlier studies noted that applying MAB methods to fraud detection was challenging due to difficulty of creating decision "arms" [12]. To address this issue, newer approaches based on Contextual Multi-Armed Bandits (CMAB) have been proposed. Some studies combine CMAB with incremental regression trees and semi-supervised learning techniques to dynamically group transactions with similar characteristics. This allows the model to continuously explore potentially suspicious patterns while still exploiting previously identified fraud indicators [13]. In addition, recent studies show that multi-agent contextual bandits offer greater efficiency in resource-constrained environments [14]. This strongly supports the use of multi-agent contextual bandits to handle upcoding within the strict audit limits of BPJS Kesehatan.

1.2. Research Gaps

Recent studies show that there are still substantial issues with the application of these technologies to a huge and a completely decentralized healthcare system such as Indonesia's BPJS Kesehatan. The paper here aims at two main research gaps.:

- Ignorance of Operational Resource Constraints: Existing high-recall ML frameworks and CMAB models optimize for fraud detection accuracy but implicitly assume an unlimited capacity to investigate suspicious claims [11], [13]. Human auditors are a strictly limited daily resource. There is a lack of literature applying Bandits with Knapsacks (BwK) framework to healthcare fraud, which is necessary to mathematically pace and optimize the allocation of limited human auditing capacity over a continuous stream of claims.
- Lack of Collaborative Multi-Agent Learning in Privacy-Restricted Environments: Current state-of-the-art fraud detection systems operate in centralized, single-agent environments. They do not address the decentralized reality of regional healthcare branch offices that must combat dynamic fraud (such as upcoding) without violating strict personal data protection laws. There is a critical gap in utilizing Federated Multi-Agent Multi-Armed Bandits (MAMAB) to allow distributed agents to collaboratively share learned adversarial fraud patterns (parameter weights) without sharing raw, sensitive patient contexts.

2. Method

To address the highly adaptive nature of upcoding fraud and the strict operational constraints of the BPJS Kesehatan verification process, we propose a decentralized, privacy-preserving Multi-Agent Multi-Armed Bandit (MAMAB) framework. The proposed model integrates Linear Contextual Bandits (LinUCB), Lagrangian-based Knapsack, and Federated Learning for multi-agent knowledge synchronization.

2.1. System Architecture

As shown in Figure 1, the end-to-end design of our proposed system runs through three main functional areas, enabling continuous real-time decision-making and periodic global learning. The initial point is in the Data Pipeline and Context Extraction section. The system gets raw claim documents from the BPJS Claim Stream at the hospital providers, and this happens before the payment is made. Context extraction is done by a dedicated data engineering pipeline which converts characterization of raw categorical and numerical features into a numeric standard form vector called Context Vector x_t. The stream is then divided and sent according to the office branch code.

Once a claim is routed, it will move into the MAMAB Local Agents area where each BPJS regional office is viewed as an autonomous agent. Upon receiving the context vector x_t, the local agent decides whether to call the claim for audit or give it an auto-approval based on the agent's local policy $\pi(\alpha|x)$ against its strictly defined local human-auditor budget B_{local}. The agent then outputs an action $\alpha_t \in 0,1$, where 0 represents the decision to approve the claim and 1 represents the decision to audit it. This central zone also contains BPJS Headquarters (Central Server) which is responsible for the federated learning process by computing and disseminating the global fraud blueprint θ_{global} without ever requiring or storing raw patient data.

The selected action instantaneously activates the Feedback and Reward Mechanism zone. If the agent decides on $\alpha_t = 0$, the claim is routed to auto-approve, requiring no human intervention. The system instantly generates a feedback loop of $(r_t = 0,\ c_t = 0)$ back to the agent. On the other hand, if the agent decides on $\alpha_t = 1$, the claim is routed to a human auditor who investigates the case and provides a ground truth label $y_t \in 0,\ 1$, where 1 denotes confirmed fraud and 0 denotes a clean claim. After that, the system mathematically calculates the financial reward of the detection $(r_t = y_t \times v_t)$ where v_t represents the monetary value of the submitted claim and consumes the local audit cost $(c_t = 1)$. In this way, a full feedback tuple is produced (r_t, c_t, y_t) and sent back to the branch office. Upon receiving this feedback, the local agents update their internal mathematical models and eventually upload these parameters to the central server, allowing the entire network to adapt to newly discovered fraud patterns.

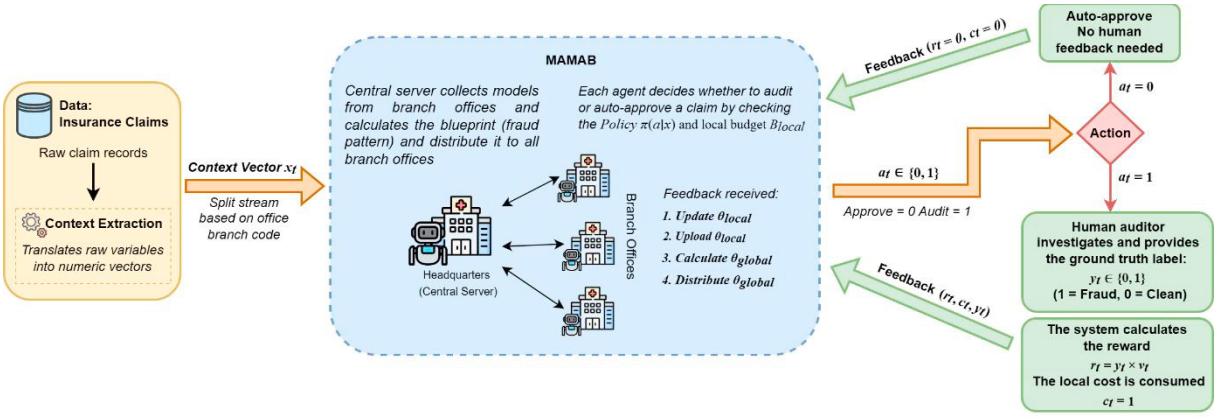


Fig. 1. Proposed system architecture of MAMAB.

2.2. Contextual Bandits with Knapsack (cBwK)

We formulate the daily claim verification process at each branch office as a finite-horizon cBwK problem defined by the tuple $T,\ A,\ X,\ D,\ B$. For a given day, the branch office receives a sequence of T claims. At each time steps $t \in |T|$, the local agent observes a context representing the claim feature vector x_t\in$X \subseteq \mathbb{R}^d$. Based on this observed context, the agent must select an action α_t from the binary action space $A = 0,1$. Selecting $\alpha_t = 0$ designates the Null Arm (Auto-Approve), which safely processes the claim without consuming any human audit resources. Conversely, selecting $\alpha_t = 1$ designates the Audit Arm, which routes the claim to a human verifier.

Each selected action incurs a resource consumption cost $c_t(\alpha_t) = \alpha_t$. This means that an automated approval costs zero and a manual audit consumes exactly one human audit slot. The agent must ensure that the cumulative cost does not exceed this limit $\sum_{t=1}^{T} c_t(\alpha_t) \leq B_{local}$. If the local audit budget is completely depleted, the agent is strictly forced to pull the Null Arm $(\alpha_t = 0)$ for the remainder of the time horizon T. Finally, the objective of the agent is driven by the reward function $r_t(\alpha_t)$, which quantifies the actual financial value of the fraud successfully prevented, is calculated as $r_t(\alpha_t) = \alpha_t \times y_t \times v_t$.

2.3. LinUCB

To avoid prematurely exhausting the budget due to an excessive number of checks, local agents must balance the probability of detecting fraud against the cost of deploying a human inspector. We model this decision-making process

using the Linear Contextual Bandits (LinUCB) algorithm [7]. As defined in the linear contextual bandits framework, the agent assumes that the expected reward in the event of fraud $\mu(\alpha|x_t)$ for an action α given a claim context x_t is a linear function of an unknown parameter vector θ_α [7].

$$\mu(\alpha|x_t) = x_t \cdot \theta_\alpha \tag{1}$$

For the Null Arm, the expected reward is known to be exactly zero $\mu(0|x_t) = 0$. For the Audit Arm $\alpha = 1$, the agent maintains a "confidence region" C_t that contains the true fraud parameter θ_1 with high probability [7]. For an incoming claim x_t, the agent calculates the Upper Confidence Bound (UCB) of the expected fraud reward by finding the parameter θ within the confidence region C_t that maximizes the expected outcome:

$$UCB_t(1|x_t) = sup_{\theta \epsilon C_t} x_t \cdot \theta \tag{2}$$

To incorporate the local auditing capacity constraint, we adapt the LagrangeBwK approach [7]. We introduce a Lagrangian multiplier $\lambda \geq 0$, which acts as a dynamic "shadow price" representing the scarcity of the remaining budget. The agent evaluates the penalized score for auditing.

$$Score_t(1) = UCB_t(1|x_t) - \lambda c_t(1) \tag{3}$$

Therefore, the agent's decision rule at step t is to pull the Audit Arm only if the UCB score exceeds the shadow price of the budget, provided capacity remains:

$$\alpha_t = \begin{cases} 1 \;\; if \; UCB_t(1|x_t) > \lambda \; and \; \sum_{t=1}^{T} c_t(\alpha_t) \leq B_{local} \\ 0 \;\; otherwise \end{cases} \tag{4}$$

2.4. Decentralized Federated Learning

To strictly adhere to Indonesia's Personal Data Protection Law (UU PDP), raw patient contexts (x_t) and ground-truth labels (y_t) are legally prohibited from being shared across regional branches. To overcome this limitation, we implement a decentralized Federated MAMAB protocol that enables fraud patterns to be shared globally while raw data learning remains entirely localized. Instead of transmitting sensitive patient records, the agents solely exchange the mathematical parameters that define their local confidence regions.

The learning cycle begins when a branch agent i audits a claim $\alpha = 1$ and receives ground-truth feedback comprising the reward, cost, and label tuple (r_t, c_t, y_t). Using this feedback, the agent independently updates the mathematical boundaries of its local confidence region $(C_{local,i})$ to reflect the newly observed fraud pattern. After the end of the synchronization period, each branch securely uploads to the BPJS Headquarters the parameters describing its locally updated confidence region $(\Delta C_{local,i})$. The Central Server then gathers the parameters from every M participating branch and merges them into one highly focused global model C_{global}. Finally, the Central Server broadcasts the updated global parameters back to all local agents. With this federated approach, when one branch discovers a new upcoding pattern, the system updates its confidence region around the related context vector. As a result, other branches can detect similar fraudulent contexts in the next operational cycle without sharing raw patient data.

3. Results

Since this paper is intended as a preliminary study, empirical simulations are still ongoing but not yet done. Nevertheless, the proposed Multi-Agent Multi-Armed Bandit (MAMAB) framework already provides several analytical insights for the BPJS Kesehatan system through its mathematical formulation and overall design.

- Optimal Utilization of Limited Audit Resources: Traditional greedy algorithms deplete auditing capacity early, leaving systems vulnerable to late-day adversarial attacks. By integrating a Lagrangian multiplier λ into the

LinUCB scoring function, our framework converts the hard constraint of human auditor capacity (B_{local}) into a dynamic shadow price. This mathematically guarantees budget pacing, ensuring a continuous defense across the entire horizon T.

- Privacy-Preserving Collaborative Adaptation: Sharing raw patient data across branches violates Indonesia's Personal Data Protection Law (UU PDP). Our Federated MAMAB architecture ensures that only non-sensitive, aggregated parameter updates are transmitted to the central server. The resulting global policy (θ_{global}) legally and mathematically protects all provincial branches against novel upcoding tactics discovered anywhere in the network.
- Value-Driven and Adaptive Fraud Detection: Traditional models optimize raw accuracy and can be misleading in highly imbalanced datasets. By defining the reward function as the actual financial value of prevented fraud, the algorithm acts as a cost-sensitive classifier. This is a theoretical proof that the system prioritizes the most expensive high-risk claims. As a result, minimizing the financial loss for BPJS Kesehatan.

4. Discussion

4.1. Contextual Bandits vs. Supervised Learning

Most automated fraud detection systems in the healthcare and financial sectors rely on supervised machine learning (ML) algorithms like Random Forest, LightGBM and deep learning architectures [10]. These models offer high precision on known patterns but their reliance on historical pre-labeled data exposes a fundamental vulnerability since they assume a static data distribution.

Healthcare fraud is highly adversarial and able to cause concept drift [12,13]. When static supervised models are deployed, they fail to recognize newly invented upcoding schemes because those schemes do not exist in the training data.

Contextual Bandits address this issue by balancing exploration and exploitation [7]. Rather than relying only on previously known fraud patterns, algorithms such as LinUCB also evaluate unfamiliar claim combinations by assigning them an uncertainty bonus. As a result, the system occasionally investigates suspicious billing patterns that have not been seen before. This allows the model to adapt more quickly to evolving fraud strategies rather than waiting for periodic software updates [12].

4.2. Single Agent vs. Multi-Agent Frameworks

Single-agent bandit architectures possess distinct advantages, primarily their computational simplicity, zero communication overhead, and absolute data privacy, as no information ever leaves the local node [15]. However, in a nationwide network like BPJS Kesehatan, their critical weakness is isolated learning. If BPJS deployed an independent single-agent Contextual Bandit at each regional branch, each node would have to "pay the cost" of exploring new fraud schemes. Recent studies on federated bandits demonstrate that in the absence of information exchange with neighbors, an isolated agent can suffer a severe linear cumulative regret of $O(T)$ [16].

Our proposed MAMAB architecture overcomes this limitation through collaborative intelligence. Literature demonstrates that even a minimal level of collaboration among different agents enables a significant reduction in per-agent regret compared to a scenario where agents do not communicate [15]. Furthermore, deploying multi-agent contextual bandits has been proven to produce faster convergence and greater adaptability than single-agent baselines in complex environments [14]. However, sharing raw patient data across branches violates Indonesia's Personal Data Protection Law (UU PDP).

By integrating Federated Learning into the MAMAB framework, we achieve the mathematical benefits of multi-agent collaboration without the privacy risks [17]. Because the agents only transmit their locally updated parameter weights θ_{local} to the central BPJS server, the "exploration cost" is distributed across the entire national network. Once a single branch identifies a new upcoding vector, the aggregated global policy θ_{global} instantly immunizes all other branches against the same adversarial strategy.

4.3. Addressing Asymmetric Costs and Class Imbalance

A significant disadvantage of the standard supervised learning for fraud detection is how it cannot deal with severe class imbalance. In national health insurance system, fraudulent claims are far less common than legitimate ones. Thus, traditional algorithms try to achieve high accuracy by classifying almost all transactions as legitimate and completely miss the rare but financially devastating fraud cases [10]. Furthermore, all prediction errors are treated as equally important in standard classification models. However, the consequences in health care finance are not evenly distributed. Failing to detect a large upcoding scheme may cost much more than temporarily sending a legitimate claim for manual review [11].

Our proposed MAMAB architecture mathematically resolves both class imbalance and asymmetric costs by abandoning standard accuracy metrics in favor of value-driven reward maximization. In our Contextual Bandit formulation, the algorithm does not merely ask, "Is this claim fraud?" Instead, the reward function evaluates the actual monetary value of the claim ($r_t = y_t \times v_t$). This inherently transforms the system into a cost-sensitive learning framework. The agent learns to prioritize the auditing of high-value, high-risk claims where the expected financial recovery outweighs the operational cost of the human auditor. By explicitly aligning the algorithm's remuneration mechanism with actual financial risk, the MAMAB environment naturally addresses statistical imbalances and directly safeguards public health budgets.

If this framework is implemented, BPJS Kesehatan will be able to quickly detect cases of fraud and optimize the use of its limited audit resources while ensuring new fraud patterns are shared with all regional branches without violating patient data privacy regulations.

4.4. Future Directions and Conclusion

BPJS Kesehatan faces threats from fraud that never stops evolving. Sadly, regular audit systems can't keep up with this change, limited by their resources and tight privacy regulations. Consequently, we suggest MAMAB to address the problem. In addition, the system can exchange info on newly identified fraud patterns without compromising patients' underlying data. By combining all the above, our creation tackles a substantial problem with extreme real-life applicability.

As this paper implies a preliminary architectural study, empirical simulation remains in progress but has not yet been finished. Future research will focus on empirical validation using real-world datasets with the aim of evaluating the accuracy of fraud detection and identifying the system's limitations. Additionally, future research aims to expand the MAMAB system to cover other fraud cases in the Indonesian healthcare system while assessing the scalability potential of the decentralized synchronization process.

Data Availability

No data was used for the research described in this article. The study focuses on the theoretical design and architectural proposition of a Multi-Agent Multi-Armed Bandit framework. Details regarding the required dataset schema and simulations are reserved for future work.

Author Contribution Statement

The study was designed and written by **E. V. Nilasari** and **B. Soelistijanto**. All authors read and approved the manuscript.

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